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Development of an Integrated System for Acoustic Calibration of Stethoscopes

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Abstract

Stethoscopes may undergo material degradation that alters acoustic transmission and attenuates clinically relevant cardiopulmonary sounds. This project aimed to develop an integrated acoustic calibration system for characterizing stethoscope frequency response and signal fidelity, generating controlled signals across the 20–2000 Hz cardiopulmonary bandwidth, acquiring high-fidelity audio, and supporting repeatable spectral analysis.

Controlled sinusoidal tests demonstrated capture of 20 Hz and 50 Hz signals, with concentrated spectral energy and generally stable amplitude envelopes, although amplitude surges and noise-floor changes were observed. At 200 Hz, the spectrogram showed harmonic components at integer multiples of the fundamental frequency, indicating total harmonic distortion. An 1111 Hz test produced a continuous spectral band without attenuation or digital blindness. The optimized acquisition rate remained well above the 4000 Hz Nyquist threshold required for the 2000 Hz bandwidth.

The developed system demonstrated functional low- and high-frequency acquisition across the intended cardiopulmonary range. However, observed gain-related amplitude variations and harmonic distortion indicate that further refinement is required before complete stethoscope calibration performance can be established.

Keywords: Stethoscope Calibration, Acoustic Transducer, Digital Signal Processing

Resumo

Os estetoscópios podem sofrer degradação do material que altera a transmissão acústica e atenua os sons cardiopulmonares clinicamente relevantes. Este projeto teve como objetivo desenvolver um sistema integrado de calibração acústica para caracterizar a resposta em frequência e a fidelidade do sinal do estetoscópio, gerando sinais controlados na largura de banda cardiopulmonar de 20–2000 Hz, adquirindo áudio de alta fidelidade e permitindo uma análise espectral repetível.

Testes sinusoidais controlados demonstraram a captura de sinais de 20 Hz e 50 Hz, com energia espectral concentrada e envelopes de amplitude geralmente estáveis, embora tenham sido observados picos de amplitude e variações no ruído de fundo. A 200 Hz, o espectrograma revelou componentes harmónicos em múltiplos inteiros da frequência fundamental, indicando distorção harmónica total. Um teste a 1111 Hz produziu uma banda espectral contínua sem atenuação nem cegueira digital. A taxa de aquisição otimizada manteve-se bem acima do limiar de Nyquist de 4000 Hz exigido para a largura de banda de 2000 Hz.

O sistema desenvolvido demonstrou aquisição funcional de baixas e altas frequências em toda a gama cardiopulmonar pretendida. No entanto, as variações de amplitude relacionadas com o ganho e a distorção harmónica observadas indicam que é necessário um maior aperfeiçoamento antes de se poder estabelecer um desempenho completo de calibração do estetoscópio.

Palavras-Chave: Calibração de estetoscópios, transdutor acústico, processamento digital de sinais

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List of Symbols

Symbol	Description	Units
f	frequência	Hz
f	força	N
f_s	frequência de amostragem	Hz
i	corrente	A
M	massa	kg
P	potência	W
t	tempo	s
V	tensão elétrica	V
x	deslocamento	m
θ	deslocamento angular	rad
ω	velocidade angular	$\mathrm{rad\,s^{-1}}$

Chapter 1

Introduction

1.1 State of the Art/Contextualization

Stethoscopes remain the most fundamental and prevasive diagnostic tools in clinical practice for the auscultation of cardiopulmonary and vascular sounds. Currently, traditional acoustic stethoscopes dominate over 90% of the global market due to their high durability, low cost, and independence from electrical power sources [1]. However, purely acoustic auscultation is inherently limited not only by the fallibility of human auditory perception but also by the unrecognized physical degradation of the instrument over time. Studies have demonstrated that general practitioners miss up to 57% of significant valvular heart diseases when relying on standard stethoscopes [2].

While frequently attributed to human auditory limitations, this high error rate may be significantly composed by unnoticed material failures; such as compromised acoustic seals in the eartips, hardened tubing, or worn diaphragms. These physical degradations silently alter the stethoscope's frequency response, attenuating vital low-frequency biological sounds before they ever reach the clinician's ear. To mitigate these human diagnostic errors, the medical field is experiencing the gradual emergence of electronic stethoscopes designed to amplify, filter, and digitize biological sounds.

The clinical consequences of an uncalibrated or degraded stethoscope are severe. When physical degradation, such as hardened acoustic tubing or compromised eartip seals, artificially attenuates low-frequency biological signals, critical acoustic

biomarkers like the S3 gallop (a primary indicator of heart failure for older adults) may fall below the threshold of hearing. This mech nical attenuation directly leads to false-negative diagnoses, delayed medical interventions, and compromised patient safety. Therefore, establishing a reliable calibration system is not merely a matter of instrument maintenance, but a critical requirement for ensuring diagnostic integrity.

The practice of clinical auscultation, listening to the internal biological sounds of the body, underwent a paradigm shift in 1816 with the invention of the first stethoscope by French physician René Laennec [3]. Prior to this invention, physicians relied on immediate auscultation, which required placing the ear directly against the patient's chest. This method was frequently ineffective for obese patients and posed significant social and hygienic discomforts, particularly when examining female patients. To overcome this, Laennec rolled a sheet of paper into a cylinder to channel the sound from the patient's chest to his ear, discovering that the acoustic waves were significantly amplified. He subsequently manufactured the first formal stethoscope from a hollow wooden tube, establishing the era of mediate auscultation [3].

While Laennec's original design was monaural (transmitting sound to only one ear), the instrument's architecture evolved drastically over the 19th and 20th centuries to maximize acoustic coupling. In 1851, Arthur Leared invented the first binaural stethoscope, which was quickly refined and commercialized by George Cammann [4]. This transition from a rigid wooden cylinder to flexible tubing connected to two earpieces allowed physicians to utilize stereo hearing, significantly improving the isolation of subtle cardiovascular sounds from ambient room noise.

The most critical advancement in the modern stethoscope's acoustic design occurred in the mid 20th century with the work of Rappaport and Sprague, followed by David Littmann. They engineered the dual-head chestpiece, incorporating a hollow bell for capturing low-frequency sounds (such as the S3 and S4 gallops) and a stiff diaphragm for high-frequency sounds (such as pulmonary crackles) [4]. While Littmann's subsequent innovations in lightweight materials and high-density acoustic tubing finalized the anatomical design that dominates modern clinics, the fundamental reliance on the mechanical transmission of acoustic pressure has remained unchanged. It is this exact reliance on physical materials that makes the modern stethoscope so susceptible to the mechanical degradation and acoustic attenuation addressed in this project.

1.2 Project Description

This project proposes the development of an acoustic calibration system designed to characterize the frequency response, and signal fidelity of stethoscopes. The proposed architecture consists of two primary domains: a physical hardware

module for sound generation and detection, and a software platform for control, data acquisition, and spectral analysis.

1.2.1 Objectives

By the end of the project, the developed system should be able to:

- **Simulate Cardiopulmonary Acoustics:** Generate controlled acoustic signals that mimic the frequency spectrum of biological sounds, particularly in the frequency range for cardiopulmonary auscultation.
- **Capture High-Fidelity Audio Data:** Accurately measure the stethoscope's output signal using a high-fidelity microphone and data acquisition system, ensuring that the measurement setup itself does not introduce significant distortions.
- **Analyze Frequency Response:** Analyze the acquired audio data to compute the stethoscope's frequency response curve utilizing python-based spectral analysis techniques.
- **Establish a Repeatable Protocol:** Develop a standardized testing protocol that can be easily replicated for routine stethoscope calibration in clinical settings, ensuring consistent and reliable performance assessments.

Chapter 2

Literature Review

While traditional acoustic stethoscopes remain the primary diagnostic tool in clinical practice, extensive literature demonstrates their susceptibility to mechanical degradation over time. Research into medical-grade polyvinyl chloride (PVC) polymers indicates that frequent exposure to standard clinical sanitizing agents, specifically isopropyl alcohol, accelerates the leaching of essential plasticizers from the polymer matrix [5]. Furthermore, studies evaluating material fatigue in clinical environments establish that the continuous absorption of lipophilic fluids, such as human skin sebum, directly contributes to polymer hardening and surface microcracking [6]. While the chemical degradation of medical PVC is well-documented in polymer science, its direct correlation with acoustic transmission loss in diagnostic instruments remains underexplored.

The structural integrity of the PVC tubing is the primary variable in preserving the acoustic pressure transmission of a stethoscope. Ertel et al. [7] established the foundational acoustics of stethoscope transmission, demonstrating that compromised structural integrity actively attenuates low-frequency biological signals. As the polymer hardens and loses its mechanical compliance, the tube's ability to reject external ambient noise is severely diminished. Leng et al. [8] highlighted that this physical degradation in the acoustic pathway drastically lowers the signal-to-noise ratio, effectively masking faint physiological frequencies.

Despite these recognized diagnostic risks, there is a distinct lack of standardized, objective calibration frameworks designed to quantify this acoustic loss. While Klum

et al. [9] proposed a validation system specifically for digital stethoscopes, a gap remains for a low-cost, automated calibration rig capable of continuously tracking the frequency response of traditional acoustic models across the 20-2000 Hz spectrum.

Recent advancements in medical diagnostics have driven a gradual transition from traditional acoustic stethoscopes to electronic variants. Electronic stethoscopes utilize digital amplification and active filtering to enhance clinical sensitivity. Studies by Fan et al. [10] demonstrated that electronic tele-auscultation systems exhibit higher sensitivity in detecting subtle cardiovascular murmurs and pathological gallops compared to unamplified acoustic instruments. Despite these technological advantages, traditional acoustic stethoscopes remain the dominant clinical standard worldwide due to their durability, lack of power requirements, and significantly lower cost. Because the global healthcare infrastructure continues to rely heavily on these purely mechanical devices, establishing a standardized calibration rig to verify their acoustic fidelity remains a critical engineering priority.

The accurate evaluation of cardiopulmonary acoustics requires specialized digital signal processing techniques, as biological sounds are inherently non-stationary signals. While standard Fast Fourier Transform (FFT) algorithms are effective for extracting the overall frequency components of a signal, they completely discard temporal data, making it impossible to map exactly when a specific frequency occurred. According to Debbal et al. [11], traditional FFT analysis is inadequate for distinguishing between transient cardiopulmonary events, such as the S1 and S2 heart sounds, which occur milliseconds apart. To resolve this, biomedical signal processing relies on the Short-Time Fourier Transform (STFT), which computes the Fourier transform across overlapping sliding windows. The resulting spectrogram provides a two-dimensional time-frequency representation, allowing researchers to visualize both the frequency spectrum and the precise timing of acoustic events.

Chapter 3

Methodology

3.1 System Requirements

To design an integrated calibration rig capable of characterizing acoustic and electronic stethoscopes, the system must satisfy a rigorous set of functional requirements (defining operational capabilities) and non-functional requirements (defining performance constraints and signal fidelity).

3.1.1 Functional Requirements

The functional requirements outline the necessary signal generation, mechanical coupling, and digital processing capabilities of the calibration rig, as detailed in Table 3.1.

Table 3.1: Functional requirements for the integrated system.

Requirement	**Engineering Description**
Controlled Acoustic Excitation	The system must generate continuous logarithmic sine sweeps covering the full cardiopulmonary frequency spectrum (20 Hz to 2000 Hz).
Structure-Borne Vibration	Acoustic energy must be delivered via a surface exciter mechanically coupled to an elastomeric tissue-simulating medium, avoiding free-air acoustic impedance mismatch.
Acoustic Sealing & Acquisition	The acquisition sensor must form an airtight seal with the stethoscope eartip to prevent low-frequency attenuation and accurately capture cardiopulmonary acoustics down to 20 Hz.
Signal Pre-Amplification	The low-amplitude analog output from the microphone capsule must be buffered and amplified to line-level voltages without clipping or harmonic distortion.
Digital Signal Processing (DSP)	The Central Control PC must acquire the digitized audio stream, compute the Fast Fourier Transform (FFT), and derive the system's frequency response curve.

3.1.2 Non-Functional Requirements

Non-functional requirements dictate the electronic precision, noise limits, and sampling criteria necessary to ensure clinical relevance and measurement repeatability, as presented in Table 3.2.

Table 3.2: Non-functional performance constraints and hardware specifications.

Requirement	**Engineering Description**
Low Total Harmonic Distortion	The power amplification stage must maintain high linearity across the target spectrum without introducing high-frequency switching noise, justifying a Class-AB analog topology.
Ultra-Low Input Noise Floor	The pre-amplifier stage must introduce minimal equivalent input noise ($e_n \leq 5$ nV/$\sqrt{\text{Hz}}$) to resolve faint biological sounds such as pulmonary crackles.
Nyquist-Compliant Sampling	The ADC stage must sample at a minimum rate of 44.1 kHz with at least 12-bit resolution to prevent quantization error and aliasing.
Measurement Repeatability	The physical coupler and elastomer medium must allow different stethoscope models to be tested under identical mechanical loading conditions.

3.2 System Architecture

To satisfy the functional and non-functional requirements established in Section 3.1, the calibration rig is divided into three interconnected domains: the *Sound Generation Stage*, the *Mechanical Domain*, and the *Sound Acquisition Stage*. Figure 3.1 illustrates the complete electromechanical signal path of the proposed architecture.

The signal path originates at the **Central Control PC**, where a digital reference waveform is converted to an analog signal (V_{out}) via a digital-to-analog converter (DAC). This signal drives a custom discrete Class-AB power amplifier based on the TDA2030 integrated circuit, which provides the linear current necessary to excite the Dayton Audio DAEX25 surface exciter without introducing high-frequency switching artifacts.

Within the *Mechanical Domain*, the exciter transfers structure-borne vibrations through a silicone elastomer phantom into the stethoscope's chestpiece. The acoustic wave travels through the stethoscope's internal acoustic tube to the eartip, where it is captured by a -40 dB Electret Condenser Microphone (ECM) capsule housed within a compliant physical coupler designed to ensure an airtight acoustic seal.

Finally, in the *Sound Acquisition Stage*, the microvolt-level signal produced by the ECM capsule is conditioned by an ultra-low-noise pre-amplifier circuit. The boosted line-level signal is digitized by a dedicated 12-bit USB Analog-to-Digital Converter and transmitted back to the Central Control PC for Fast Fourier Transform (FFT) and spectral analysis.

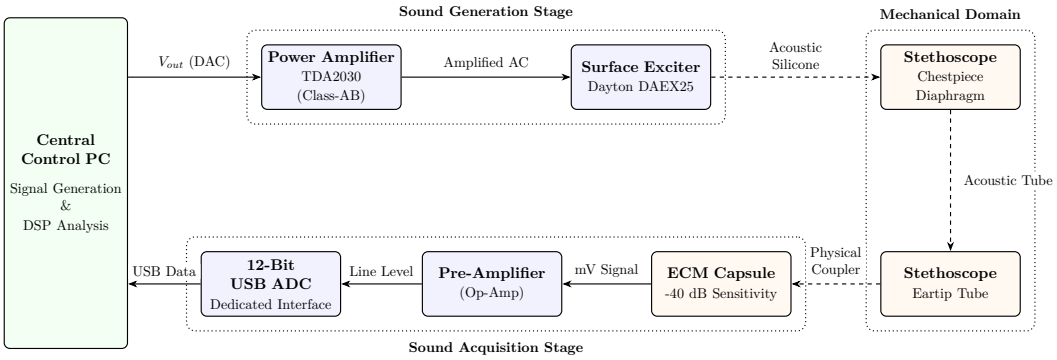


Figure 3.1: Block diagram of the proposed physical rig

3.3 Component Selection

3.3.1 Sound Generation Technologies

Traditional audio testing frequently relies on standard dynamic speakers. However, dynamic speakers operate by displacing free air.

Because a medical stethoscope is clinically designed to capture vibrations through high-impedance solid mediums (such as human tissue and fluid), utilizing an air-coupled speaker introduces significant energy loss, particularly in the low-frequency cardiovascular bands (20 Hz - 150 Hz).

To overcome this limitation, modern biomedical testing methodologies require the use of surface transducers or exciters. Instead of pushing air, surface transducers are designed to directly vibrate the solid object to which they are attached. When a surface transducer is mechanically coupled with a tissue-simulating medium (such as silicone or acoustic gelatine), it can accurately replicate the acoustic impedance of a human chest [9].

Following the justification for structure-borne sound converters, three commercially available surface transducers were selected for evaluation in this project:

- **Dayton Audio TT25-8 PUCK:** This is a compact, high-performance tactile transducer designed for low-frequency applications. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 19€.
- **Dayton Audio HDN-8 Weatherproof:** This is a high-performance, weatherproof tactile transducer designed for outdoor use. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 25€.
- **Dayton Audio DAEX25FHE-4:** This is a compact and affordable exciter with a frequency response of 70 Hz to 20 kHz and comes at a price range of around 15€.

For the sake of this project, the **Dayton Audio DAEX25FHE-4 exciter** was selected as the most suitable option for our stethoscope calibration system. This choice was based on its ability to produce strong vibrations in the relevant frequency range for cardiopulmonary sounds, as well as its affordability and compact size, which are important considerations for a practical calibration system.

Power Amplifier

To drive the Dayton Audio DAEX25FHE-4 exciter, the reference signal generated by the central control software must be passed through a power amplifier. The primary requirement for this component is signal linearity; the amplifier must deliver a perfectly flat frequency response from 20 Hz to 2000 Hz without introducing Total Harmonic Distortion (THD) or low-frequency roll-off.

Two primary amplifier topologies were evaluated for the physical test rig:

- **Class-D Amplifiers (e.g., PAM8403 / TPA3116):** Class-D modules operate utilizing PWM. While extremely affordable and capable of operating on

standard 5V USB power, the digital switching nature of Class-D topologies introduces high-frequency switching noise and poorer inherent linearity [12].

- **Class-AB Amplifiers (e.g., TDA2030 / LM1875):** Class-AB amplifier provides a continuous, highly linear analog output, ensuring a pure analog signal path across the entire target frequency spectrum. Literature demonstrates that Class-AB topologies consistently outperform Class-D stages in Total Harmonic Distortion (THD) metrics, frequently achieving distortion figures up to 30 dB better than their digital counterparts [13].

To ensure the acoustic reference signal remained free of high-frequency digital artifacts and low-frequency attenuation, the **Class-AB TDA2030** IC was selected. It comes at a highly cost-effective price point of around 2€.

The TDA2030 was then integrated into a power amplifier circuit to provide sufficient current to drive the Dayton Audio DAEX25FHE-4 exciter without introducing clipping or harmonic distortion.

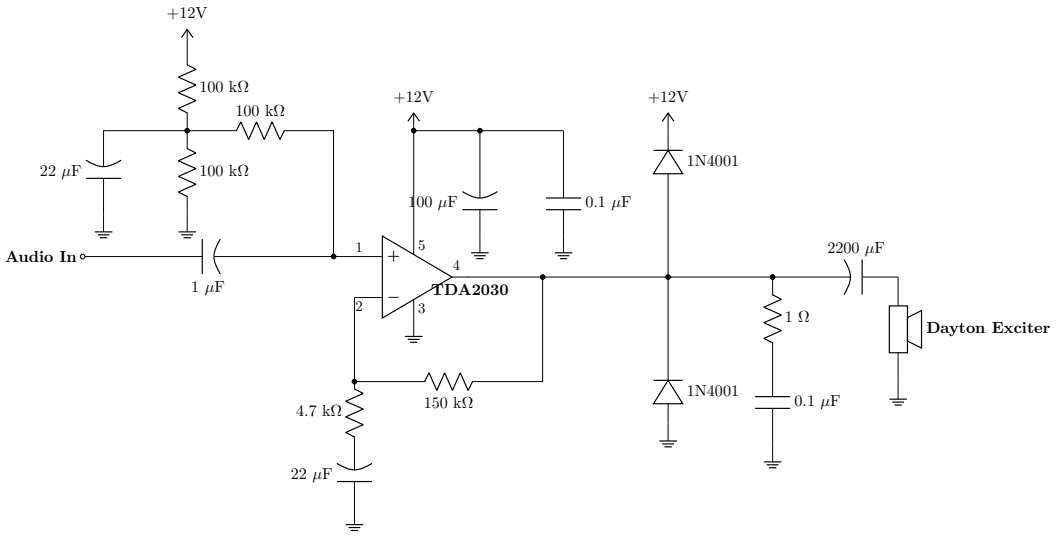


Figure 3.2: TDA2030 Module Schematic.

3.3.2 Sound Acquisition Technologies

To accurately capture the acoustic output of the stethoscope, the microphone must satisfy two critical requirements: it must possess a physical form factor small enough to achieve an airtight seal with the stethoscope's eartip (approximately 5 mm in diameter), and it must feature a flat frequency response capable of registering extreme low-frequency cardiovascular sounds down to 20 Hz.

Two primary microphone technologies were evaluated for the acquisition stage:

- **MEMS (Micro-Electro-Mechanical Systems) Microphones:** Commonly utilized in modern mobile devices, MEMS microphones are exceptionally compact and digitally integrated. However, consumer-grade MEMS capsules are specifically engineered for human speech transmission. Consequently, they frequently incorporate built-in high-pass filters designed to attenuate wind and handling noise below 60 Hz. Utilizing a standard MEMS microphone would result in the severe attenuation of foundational S1, S2, and S3 heart sounds, rendering them unsuitable for this calibration system.
- **ECM (Electret Condenser Microphone) Capsules:** ECMs are traditional analog acoustic sensors utilized in audio measurement. A standard 6 mm ECM capsule provides the necessary compact form factor for an airtight eartip coupling. More importantly, unlike consumer MEMS devices, measurement-grade ECM capsules do not artificially filter low-frequency kinetic energy, providing a highly linear frequency response down to 20 Hz.

Given the low-frequency requirements of cardiopulmonary auscultation, a 6 mm ECM capsule was selected as the optimal sensor for the acquisition stage. Because the raw analog output of an ECM capsule is highly attenuated (typically in the millivolt range), this selection necessitates the integration of a low-noise pre-amplification stage prior to digital signal processing.

Table 3.3: Comparison of microphone sensitivity and voltage output for different ECM capsules.

Microphone Example	**Sensitivity (dBV/Pa)**	**Voltage Output (mV/Pa)**
Generic Low-Cost ECM	-60 dB	1.00 mV
High-Sensitivity ECM	**-40 dB**	**10.00 mV**
Premium Measurement Mic	-20 dB	100.00 mV

As demonstrated in Table 3.3, microphone sensitivity operates on a logarithmic scale. A Generic Electret Condenser Microphone rated at -60 dB outputs approximately 1.00 millivolts per Pascal of acoustic pressure. By acquiring a higher sensitivity capsule rated at -40 dB, the electrical output increases drastically to 10 millivolts per Pascal, yielding a signal almost 10 times stronger for the exact same acoustic event.

This stronger signal reduces the gain requirements for the subsequent ADC preamplification stage, therefore minimizing the amplification of inherent electronic noise and improving the overall Signal-to-Noise Ratio (SNR) of the acquired data.

Pre-Amplifier

Despite selecting a high-sensitivity Electret Condenser Microphone (-40 dBV/Pa), the raw analog output generated by cardiopulmonary acoustics remains in the low millivolt range. To prevent quantization error and signal loss during analog-to-digital conversion, the microphone's output must be conditioned and boosted using a dedicated pre-amplifier stage.

To address the extremely low amplitude of physiological sounds, the transducer was interfaced with a MAX9814 microphone amplifier module. The MAX9814 was selected for its integrated low-noise preamplifier and Automatic Gain Control (AGC) circuitry. The AGC dynamically compresses loud ambient noises while maximizing sensitivity to faint biological signals, which is critical for unpredictable clinical environments. The module was powered via the 3.3 V rail of the microcontroller, and its gain pin was tied high to lock the amplification at 40 dB, providing an optimal balance between signal enhancement and ambient noise floor rejection.

Because cardiopulmonary acoustics are alternating current (AC) waveforms, the signal required a precise direct current (DC) bias to be accurately read by the single-supply analog-to-digital converter. The MAX9814 outputs a resting DC bias of 1.25 V. To safely center the waveform for the microcontroller, a voltage divider network was constructed using two 100 kΩ resistors between the 3.3 V supply and ground, establishing a stable 1.65 V to 1.8 V reference node.

A 10 μF electrolytic coupling capacitor was placed in series between the MAX9814 output and the voltage divider. This capacitor acts as a DC block, preventing the 1.25 V bias of the amplifier from colliding with the 1.8 V bias of the microcontroller node, while allowing the AC biological audio frequencies to pass through unimpeded.

Hardware Anti-Aliasing Filter

To prevent high-frequency environmental noise and digital aliasing from corrupting the target cardiopulmonary data, a passive resistor-capacitor (RC) low-pass filter was engineered directly at the microcontroller input. The target acoustic bandwidth for heart and lung sounds spans from 20 Hz to 2000 Hz. A 5.6 kΩ series resistor and a 10 nF ceramic capacitor were selected to construct the filter. The cutoff frequency (f_c) was calculated using the standard RC filter equation:

$$f_c = \frac{1}{2\pi RC} \tag{3.1}$$

$$f_c = \frac{1}{2\pi(5600\ \Omega)(10 \times 10^{-9}\ \text{F})} \approx 2842\ \text{Hz} \tag{3.2}$$

This 2842 Hz cutoff frequency acts as an acoustic window. It allows 100% of the 2000 Hz medical bandwidth to pass into the ADC mathematically untouched, while aggressively attenuating high-frequency room static and serial transmission noise.

3.3.3 Signal Digitization and Transmission

Following the analog signal conditioning stage, the acoustic data must be digitized for computational analysis. Signal digitization was executed using an **ESP32 microcontroller**, leveraging its internal 12-bit Analog-to-Digital Converter (ADC). This ADC resolution allows the biological waveform to be mapped across 4096 discrete voltage steps.

A critical limitation in digital auscultation is the Nyquist sampling theorem, which dictates that the sampling rate must be at least twice the highest frequency of interest to prevent data loss. To accurately visualize a 2000 Hz cardiopulmonary spectrogram, a minimum sampling rate of 4000 Hz was strictly required. Standard serial communication speeds create a data bottleneck, physically capping the sampling rate and blinding the software to the upper diagnostic frequencies. To overcome this limitation, the ESP32 serial baud rate was set to 500,000 bps, and the loop delay to 50 microseconds. This optimization successfully pushed the hardware sampling rate well above the 4000 Hz threshold, fulfilling the Nyquist criteria for the entire lung sound bandwidth.

The raw serial data was subsequently logged locally using a custom Python script utilizing the PySerial library, preparing the time-series data for spectral analysis.

Chapter 4

Results and Discussion

4.1 System Characterization via Controlled Frequency Sweeps

To evaluate the spectral response and operational limits of the integrated hardware prior to mechanical stethoscope coupling, a series of controlled sinusoidal tone sweeps were administered. The system's digital acquisition pipeline logged time-domain amplitude variations and converted the signals via Fast Fourier Transform (FFT) into dB-scaled spectrograms.

4.1.1 Low-Frequency Biomedical Bandwidth Analysis (20 - 50 Hz)

Biomarkers such as S1 and S2 heart sounds, alongside pathological S3 and S4 gallops, rely heavily on extreme low-frequency preservation. Testing the system at a fundamental frequency of 20 Hz verified the hardware's capacity to process foundational kinetic energy without artificial signal attenuation or low-frequency roll-off.

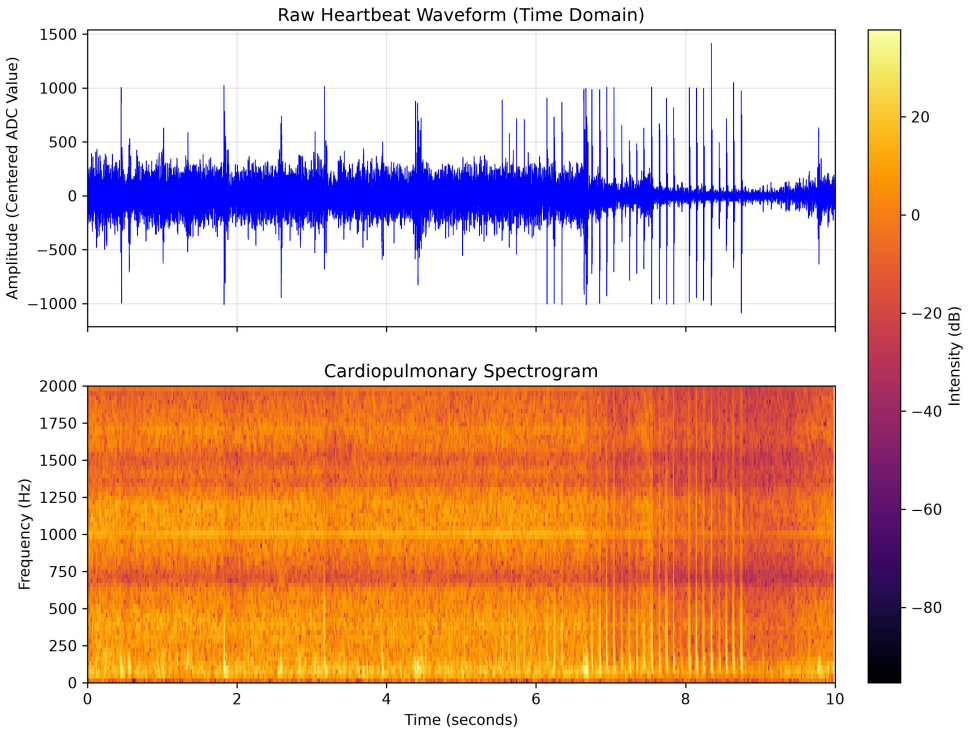


Figure 4.1: Time-domain waveform and corresponding spectrogram of a 20 Hz sinusoidal tone sweep. The waveform demonstrates clean periodicity with no DC offset, while the spectrogram confirms high-intensity energy capture at the 20 Hz frequency.

As demonstrated in Figure 4.1, the time-domain waveform while noisy, maintains a consistent amplitude envelope, and the spectrogram confirms that the system accurately captures the 20 Hz frequency with minimal harmonic distortion. However the sudden accuracy increase after 7 seconds is indicative of a noise floor reduction, likely due to the microphone's AGC (Automatic Gain Control) algorithm engaging after a period of low-amplitude input.

Extending the low-frequency evaluation to 50 Hz provides further insight into the system's tracking capabilities across the primary cardiovascular band.

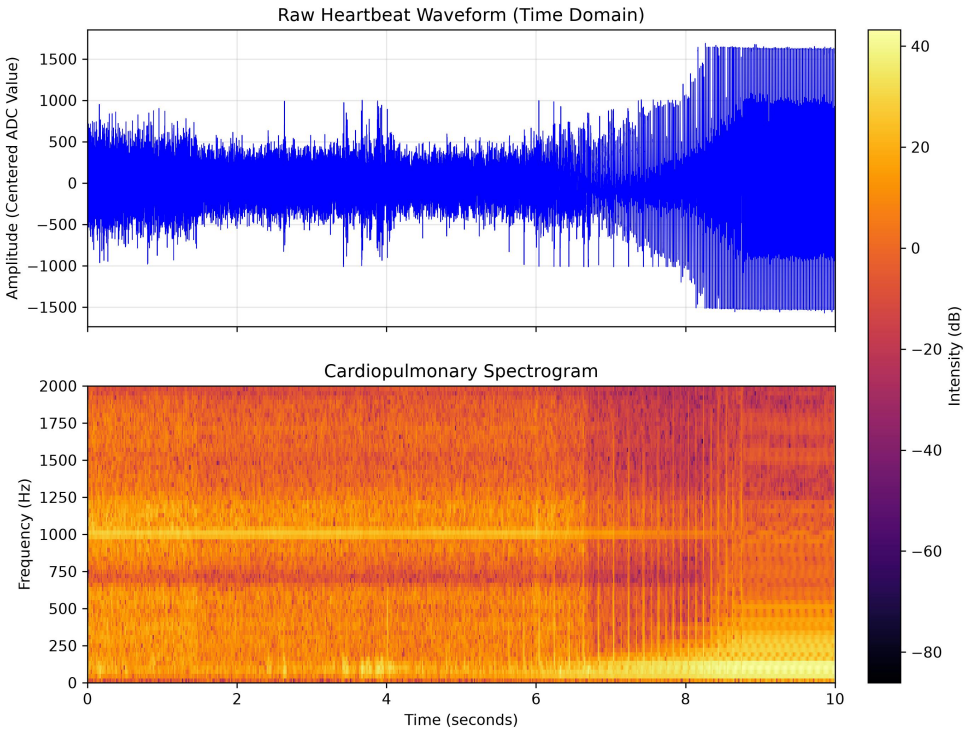


Figure 4.2: Time-domain waveform and corresponding spectrogram of a 50 Hz sinusoidal tone sweep. The waveform exhibits a stable amplitude envelope, while the spectrogram confirms a concentrated energy band at 50 Hz

The Figure 4.2 displays a dense, high-intensity band concentrated at the 50 Hz baseline. The time-domain envelope remains stable for the majority of the capture window before exhibiting a high-amplitude surge during the final two seconds. This surge pushes the waveform toward the 1500 ADC value threshold, resulting in a broader vertical dispersion of energy across the upper frequency tiers of the spectrogram.

4.1.2 Intermediate-Frequency Analysis (200 Hz)

Transitioning into intermediate frequencies, testing the system at 200 Hz evaluates the operational response and linearity of the amplification stage as the signal approaches the mid-range of the cardiopulmonary spectrum.

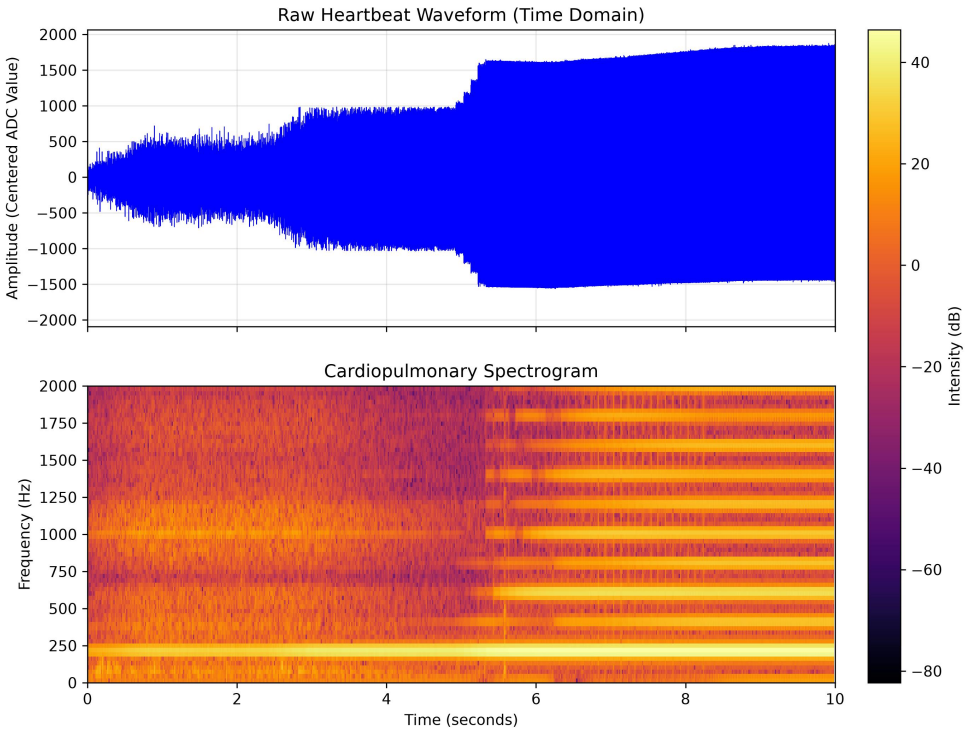


Figure 4.3: Time-domain waveform and corresponding spectrogram of a 200 Hz sinusoidal tone sweep. The waveform demonstrates a stable amplitude envelope with a sudden high-amplitude surge, while the spectrogram reveals harmonic distortion bands at integer multiples of the fundamental frequency.

As illustrated in Figure 4.3, the time-domain envelope remains relatively constrained during the first five seconds before experiencing a sharp, high-amplitude jump, expanding the waveform range toward ±1500 ADC units. Concurrently, the spectrogram exhibits a dense fundamental frequency band at 200 Hz accompanied by sharp, parallel horizontal intensity tracks stacking at exact integer multiples (400 Hz, 600 Hz, 800 Hz, 1000 Hz, and above). Rather than random system noise, these structured harmonic bands are indicative of Total Harmonic Distortion (THD).

4.1.3 High-Frequency Analysis (1111 Hz)

To verify that the digitization pipeline avoids the data bottlenecks typical of lower communication speeds and satisfies the Nyquist sampling theorem for the upper bounds of the cardiopulmonary bandwidth, high-frequency validation was performed using an 1111 Hz input tone.

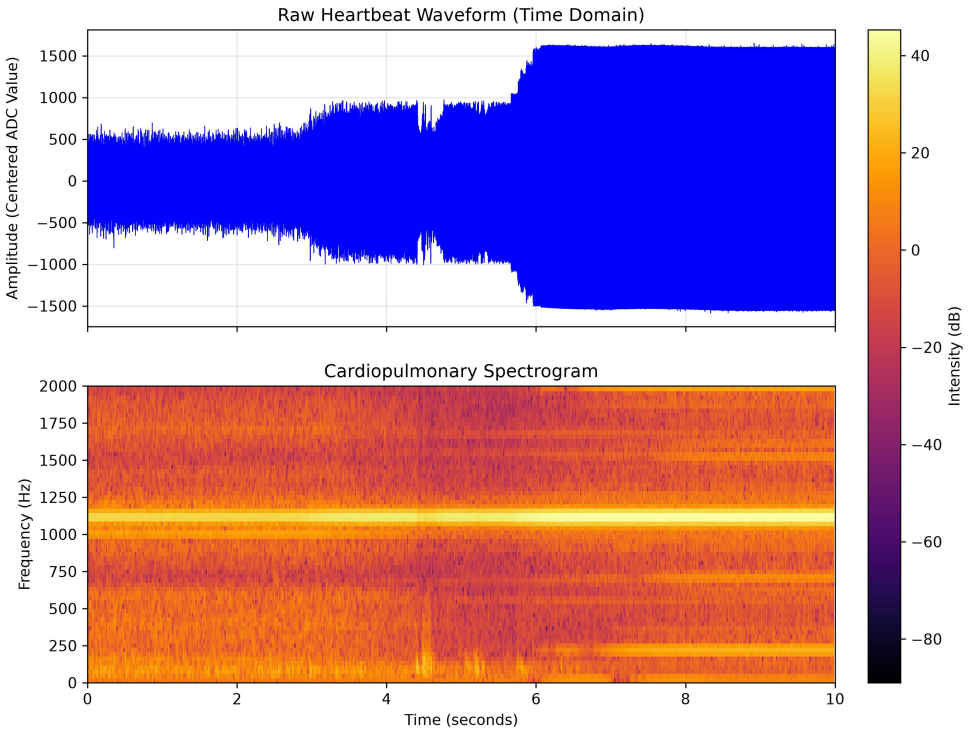


Figure 4.4: Time-domain waveform and corresponding spectrogram of an 1111 Hz sinusoidal tone test. The waveform demonstrates stable high-frequency acquisition, while the spectrogram confirms a continuous, clean energy band at 1111 Hz without attenuation or digital blindness.

The spectrogram in Figure 4.4 displays a sharp, continuous horizontal intensity band locked precisely at the 1111 Hz mark across the entire ten-second capture window. This visualization confirms that increasing the ESP32 serial communication baud rate to 500,000 bps successfully maintains a hardware sampling rate well above the 4000 Hz threshold required to map the 2000 Hz medical ceiling.

4.2 Clinical and Economic Implications of Acoustic Degradation

While the technical results confirm the system's ability to map the frequency response of a stethoscope, analyzing these findings within a broader clinical context raises critical questions regarding instrument maintenance, professional awareness, and the economic viability of acoustic versus electronic devices.

4.2.1 Clinical Awareness of Acoustic Degradation

A primary danger of stethoscope degradation is its insidious nature. The leaching of plasticizers and the hardening of PVC tubing occur over several years. Because this acoustic attenuation is incredibly gradual, the health professional's auditory perception might subconsciously adapt to the decreasing volume and clarity. Most clinicians may be entirely unaware that their diagnostic instrument has fallen below a safe clinical threshold until a significant anomaly, such as a faint S3 gallop, is entirely missed. This lack of awareness underscores the necessity of moving away from subjective auditory checks and implementing objective, routine calibration protocols in hospital environments.

In that same note, when first acquiring a stethoscope, clinicians are often unaware of the acoustic limitations of their instrument. While the initial purchase may be based on brand reputation or aesthetic appeal, the long-term acoustic performance is rarely considered, specially taking into account that there's no set timeframe for when a stethoscope should be calibrated or replaced.

4.2.2 Testing vs. Calibration and the Question of Reversibility

The development of this system inherently questions the nomenclature of "calibration" in the context of acoustic stethoscopes. In traditional electronics, calibration implies measuring a deviation and subsequently tuning the system back to a known baseline. However, the chemical breakdown of PVC and the micro-cracking of elastomeric seals are irreversible physical processes. It is chemically impossible to restore leached plasticizers to hardened tubing. Therefore, in the context of purely acoustic stethoscopes, "calibration" serves as a definitive diagnostic test to quantify acoustic loss. It establishes a mathematical threshold that dictates exactly when a clinician must physically replace the compromised components (such as purchasing replacement tubing or eartips) rather than attempting to repair the irreversible material fatigue.

4.2.3 The Economic Dilemma: Acoustic vs. Electronic Stethoscopes

Given the inevitable degradation of traditional PVC stethoscopes, the medical community faces an economic choice: continuously replace degrading acoustic parts or transition entirely to digital, electronic stethoscopes. While electronic stethoscopes resolve the issue of acoustic attenuation through active digital amplification, they introduce significant economic and logistical barriers. A standard, high-quality acoustic stethoscope costs a fraction of its digital counterpart, requires no batteries, and is highly durable. Transitioning an entire hospital's diagnostic fleet to electronic models would require massive initial capital, continuous software maintenance, and

daily charging protocols. Therefore, shifting exclusively to electronic instruments is not economically viable for the majority of global healthcare systems. A more sustainable approach is the implementation of low-cost testing rigs, like the one developed in this project, to extend the safe diagnostic lifespan of traditional acoustic instruments through targeted, data-driven component replacement.

References

[1] Transparency Market Research, "Stethoscope market - global industry analysis, size, share, growth, trends, and forecast, 2022-2031." Industry Report, 2022. [Online]. Available: https://www.transparencymarketresearch.com/. [Cited on page 1]

[2] S. K. M. Gardezi, S. G. Myerson, B. Prendergast, A. Prothero, A. Prendergast, M. Loudon, and J. D'Arcy, "Cardiac auscultation in diagnosing valvular heart disease: a comparison between general practitioners and cardiologists," *European Heart Journal*, vol. 38, no. suppl_1, 2017. [Cited on page 1]

[3] A. Roguin, "Rene theophile hyacinthe laënnec (1781–1826): the man behind the stethoscope," *Clinical medicine & research*, vol. 4, no. 3, pp. 230–235, 2006. [Cited on page 2]

[4] F. Weinberg, "The stethoscope," *Canadian Family Physician*, vol. 39, p. 2223, 1993. [Cited on page 2]

[5] M. S. Josh, S. Pradeep, S. Balachandran, R. S. Devi, *et al.*, "Temperature- and solvent-dependent migrations of di (2-ethylhexyl) phthalate, the hazardous plasticizer from commercial pvc blood storage bag," *Journal of Polymer Research*, vol. 19, no. 7, p. 9915, 2012. [Cited on page 5]

[6] Z. Zhang *et al.*, "A review of artificial sebum formulations, their compositions, uses and physicochemical characteristics," *International Journal of Cosmetic Science*, vol. 47, no. 1, pp. 168–189, 2024. [Cited on page 5]

[7] P. Y. Ertel, M. Lawrence, R. K. Brown, and A. M. Stern, "Stethoscope acoustics: Transmission and filtration patterns," *Circulation*, vol. 34, no. 5, pp. 899–909, 1966. [Cited on page 5]

[8] S. Leng, R. S. Tan, K. T. C. Chai, C. Wang, D. Ghista, and L. Zhong, "The electronic stethoscope," *BioMedical Engineering OnLine*, vol. 14, no. 1, p. 66, 2015. [Cited on page 5]

[9] M. Klum *et al.*, "Validation system for digital stethoscopes," *Current Directions in Biomedical Engineering*, vol. 5, no. 1, pp. 261–264, 2019. [Cited on pages 6 and 10]

[10] B. Fan *et al.*, "Real-world evaluation of the stemoscope electronic tele-auscultation system compared to traditional acoustic stethoscopes," *BioMedical Engineering OnLine*, vol. 21, no. 63, 2022. [Cited on page 6]

[11] S. M. Debbal and F. Bereksi-Reguig, "Computerized heart sounds analysis," in *Applied Digital Signal Processing*, IntechOpen, 2011. [Cited on page 6]

[12] Y. Zhang *et al.*, "The class d audio power amplifier: A review," *Electronics*, vol. 11, no. 19, p. 3244, 2022. [Cited on page 11]

[13] M. Duraij *et al.*, "A comparative study on class ab and class d amplifier topologies," in *2021 IEEE International Symposium on Power Line Communications and its Applications (ISPLC)*, pp. 1–6, IEEE, 2021. [Cited on page 11]